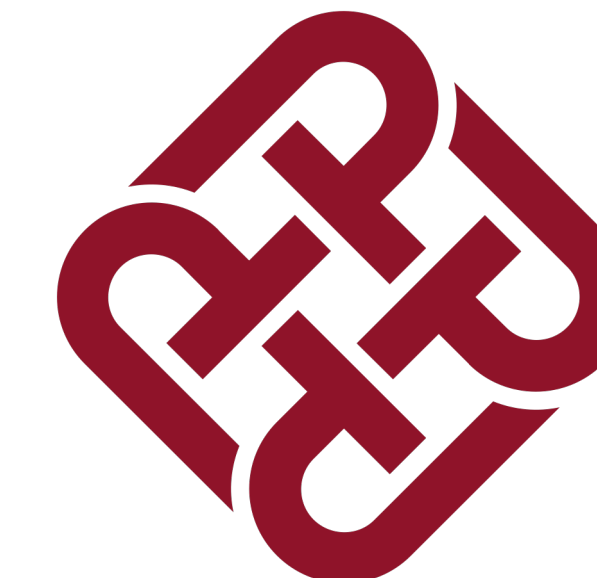




From Per-Image Low-Rank to Encoding Mismatch: Rethinking Feature Distillation in Vision Transformers

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Key takeaway:

ViT feature KD fails despite per-image compressibility: a narrow student faces an endpoint encoding mismatch from input-dependent subspace rotation and insufficient channel bandwidth.

1 Problem & Motivation

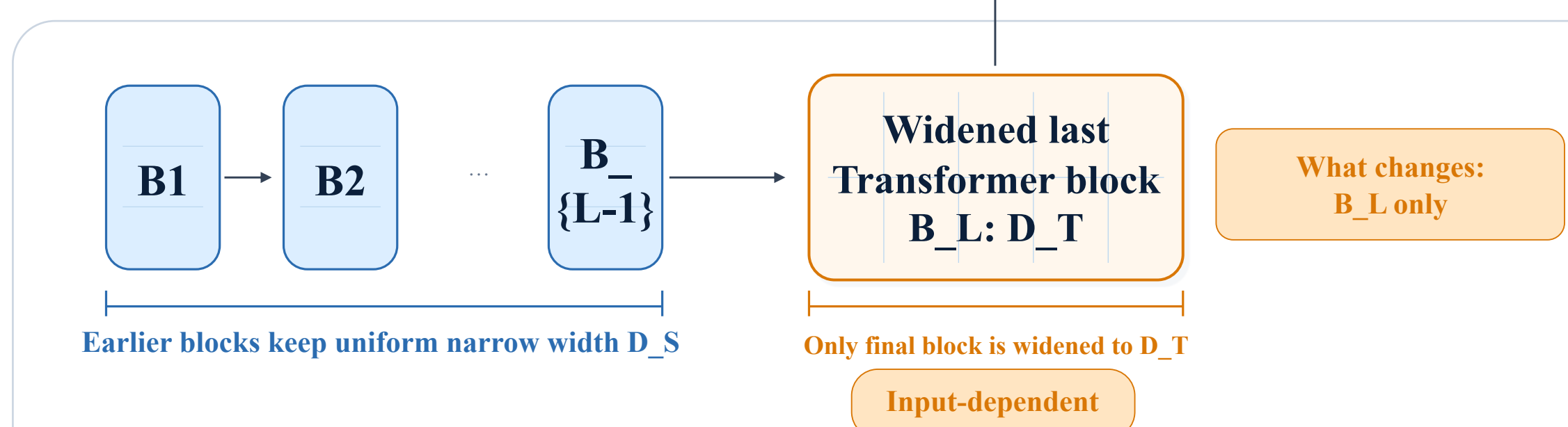
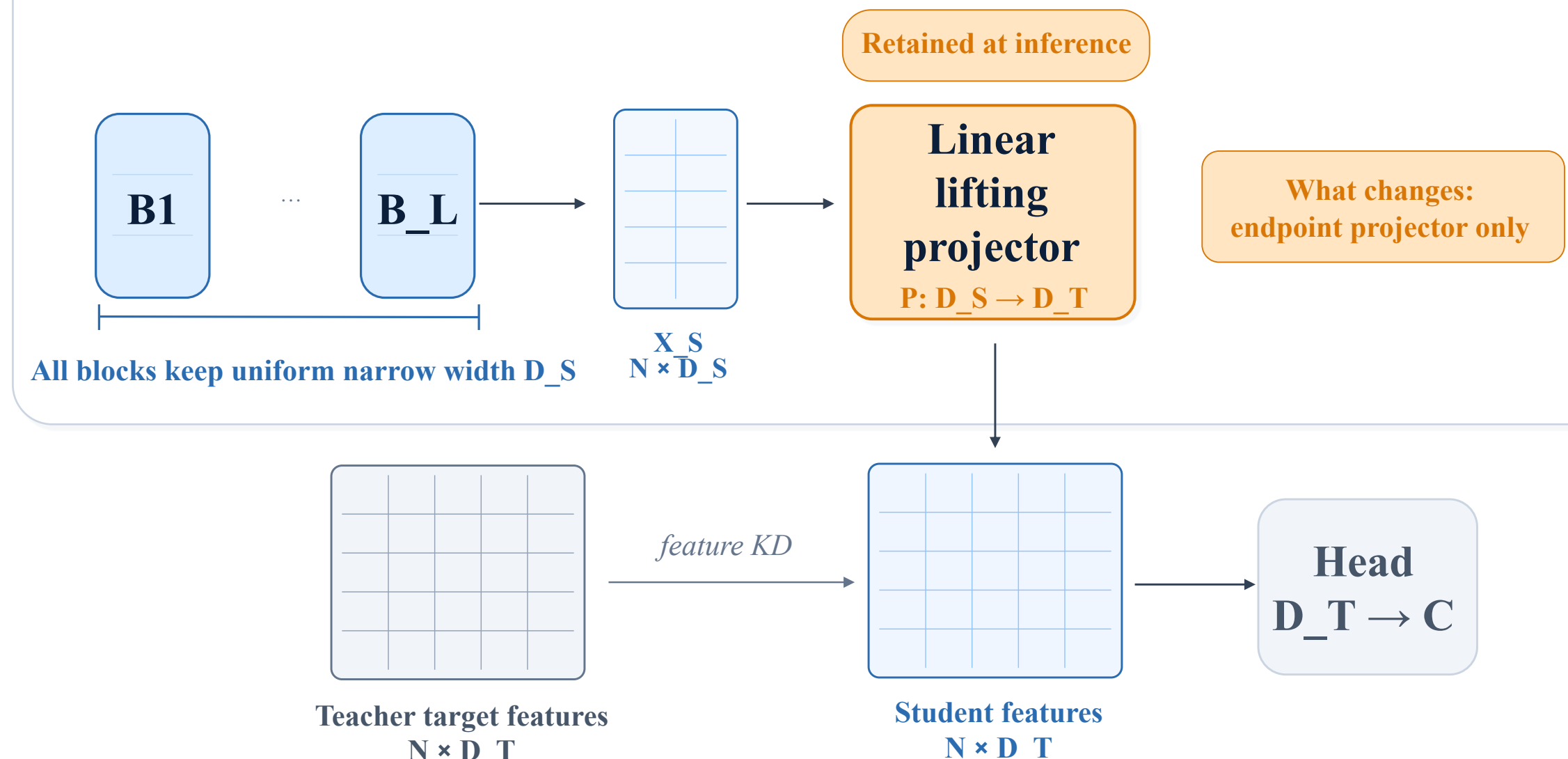
- Feature-map knowledge distillation works well in CNNs and similarly sized ViTs, but often fails in wide-to-narrow ViT compression.
- Why does simple feature matching fail even though teacher representations appear low-rank?
- We answer this through representation analysis, not heuristic engineering.

Main claim:
Interpretability first, method second.

4 Simple fixes guided by interpretability

Method 1 Lift: Post-Hoc Feature Lifting

- ✓ Add a lightweight linear lifting projector after the student's last layer.
- ✓ Retain the projector at inference.
- ✓ Mainly fixes endpoint bandwidth.

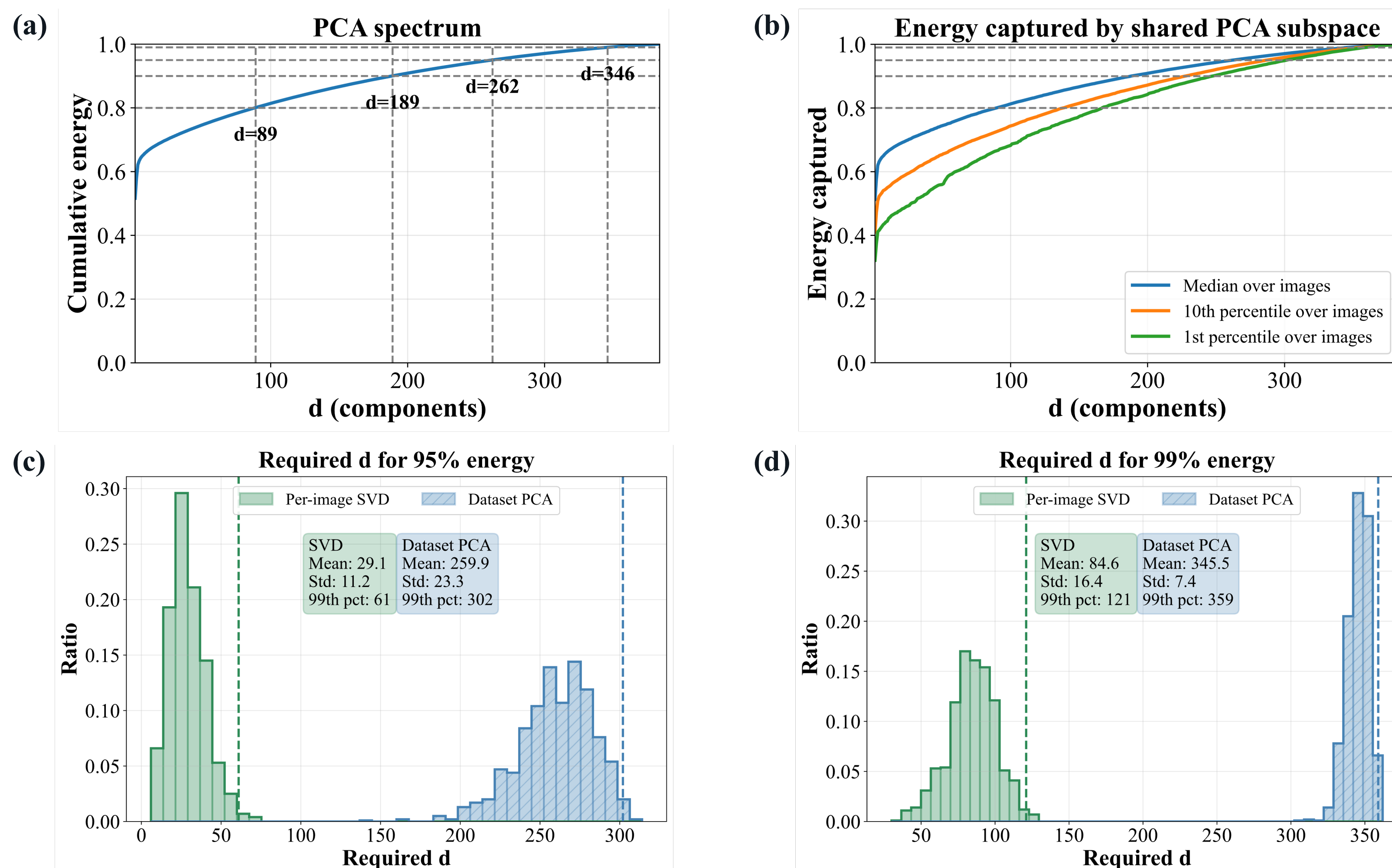


Method 2 WideLast: Native Width Alignment

- ✓ Widen only the student's final Transformer block to the teacher width.
- ✓ Provides near-teacher bandwidth plus input-dependent expansion.
- ✓ Fixes both bandwidth and orientation more strongly.

Deliberately minimal design to validate the interpretability insight.

2 Discovery 1: Per-image low-rank, but dataset-level mismatch



CaiT-S24 ($D = 384$). (a) PCA spectrum. (b) Energy captured by a shared PCA subspace. (c) Required d for 95% energy. (d) Required d for 99% energy.

- Each image is easy to compress.
- Across the dataset, the teacher behaves like a union of low-rank subspaces with strong subspace rotation.
- Therefore a single narrow linear interface is insufficient.

Key numbers (CaiT-S24, $D = 384$)

- 95% energy**
per-image SVD 99th pct. = 61
shared PCA 99th pct. = 302 / 384
- 99% energy**
per-image SVD 99th pct. = 121
shared PCA 99th pct. = 359 / 384

Facet 1: Orientation mismatch (rotation)

5 Geometry-guided fixes revive feature KD

ImageNet-1K Top-1 Acc. (%)
Teacher: CaiT-S24; Student: DeiT-Tiny

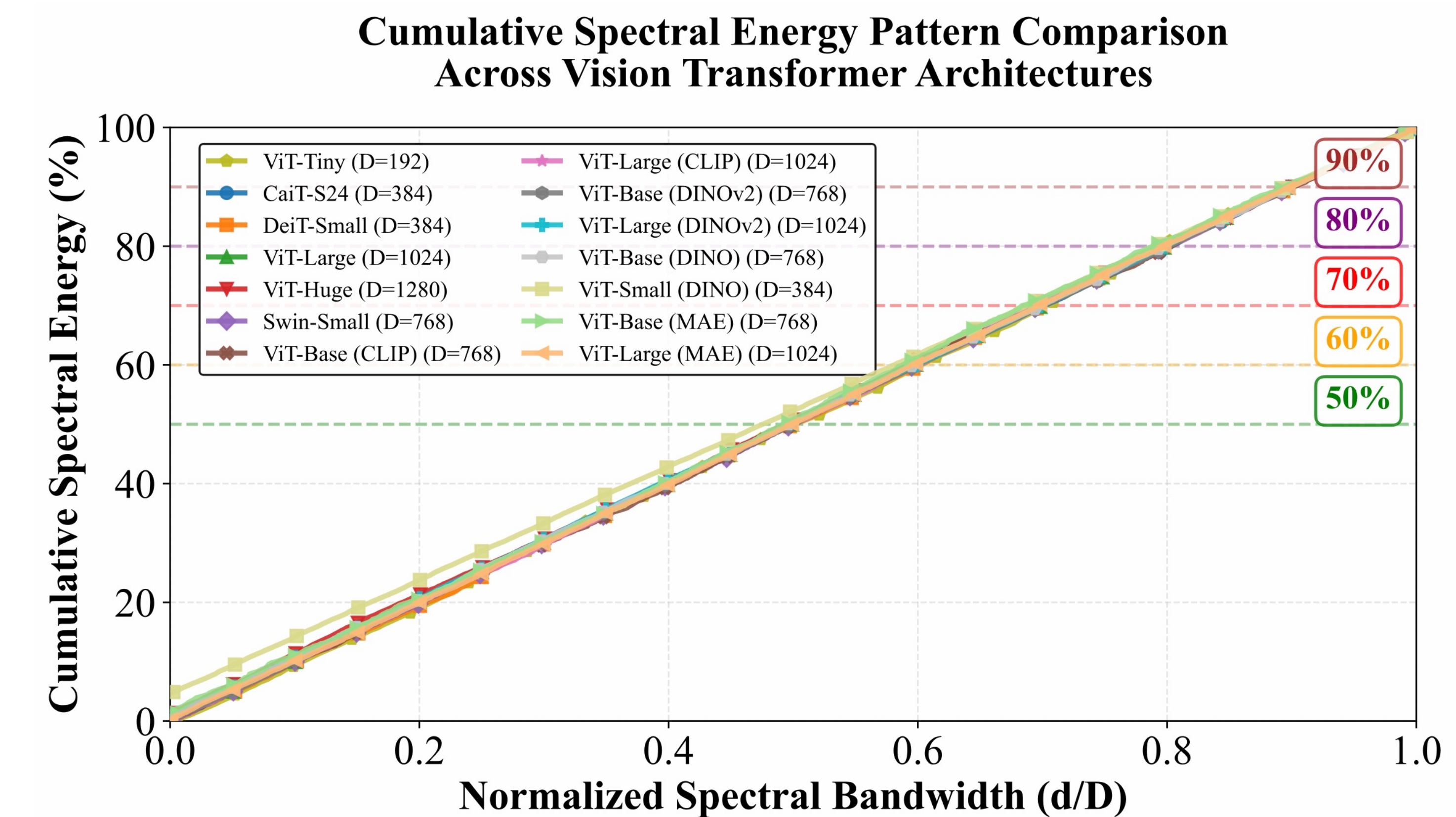
Setting	Top-1 Acc. (%)	Gain
Baseline no KD	74.86	—
Baseline + SpecKD	75.07	+0.21
Lift + SoftKD + SpecKD	77.53	+2.67
WideLast + SoftKD + MSE	78.23	+3.37

Even without distillation

Setting	Top-1 Acc.	Gain
Baseline	74.86	—
Lift	75.41	+0.55
WideLast	75.54	+0.68

Simple architecture changes improve both KD and standalone performance.

3 Discovery 2: A universal encoding law



- We introduce token-level Spectral Energy Pattern (SEP).
- Across architectures and training regimes, the normalized SEP curves almost overlap.
- This reveals an architecture-invariant encoding law: tokens spread energy broadly across channel modes.
- Even when the whole image is low-rank across tokens, each token still needs broad channel bandwidth.



Universal law: normalized spectral bandwidth required for 50 / 60 / 70 / 80 / 90% energy is approximately **0.50 / 0.60 / 0.70 / 0.80 / 0.90** across many models.

Facet 2: Capacity mismatch (bandwidth)

6 Why this matters



Explains when naive ViT feature-map KD fails.



Turns an interpretability finding into a practical design rule.



Suggests that matching endpoint encoding geometry is more important than blindly increasing distillation complexity.



Potential broader relevance: endpoint encoding geometry may matter beyond classification KD.

Interpret the representation first → Identify the encoding mismatch → Apply a minimal architectural fix → Recover performance gains.